

Lecture 5

Motion under central force

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MC2103
Fall 2026



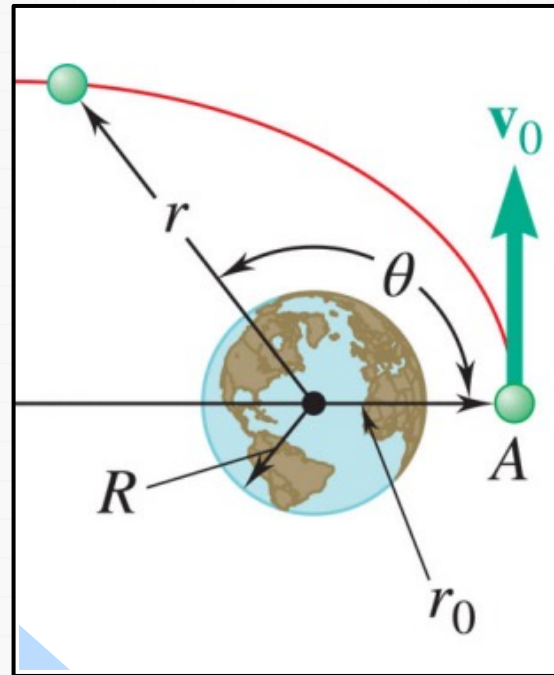
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5 Motion under central force

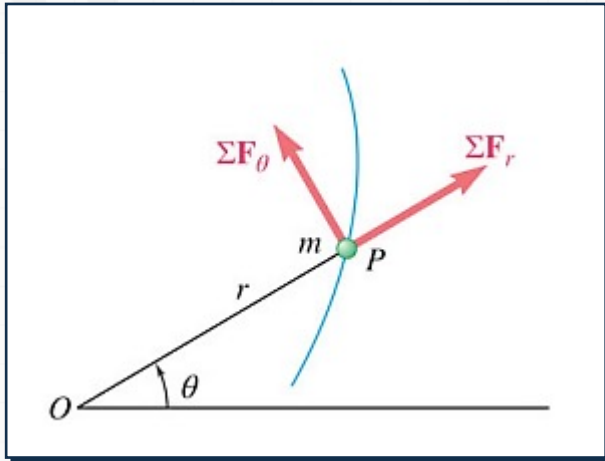
Angular Momentum of a Particle

Satellite orbits are analyzed using conservation of angular momentum.



5 Motion under central force

Eqs of Motion in Radial & Transverse Components



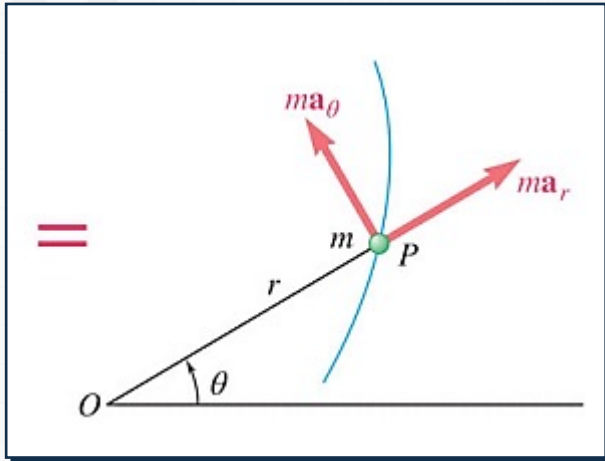
- Consider particle at r and θ , in polar coordinates,

$$\sum F_r = ma_r = m(\ddot{r} - r\dot{\theta}^2)$$

$$\sum F_\theta = ma_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

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Eqs of Motion in Radial & Transverse Components



- This result may also be derived from conservation of angular momentum,

$$\begin{aligned} H_O &= mr^2\dot{\theta} \\ r \sum F_\theta &= \frac{d}{dt}(mr^2\dot{\theta}) \\ &= m(r^2\ddot{\theta} + 2r\dot{r}\dot{\theta}) \\ \sum F_\theta &= m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) \end{aligned}$$

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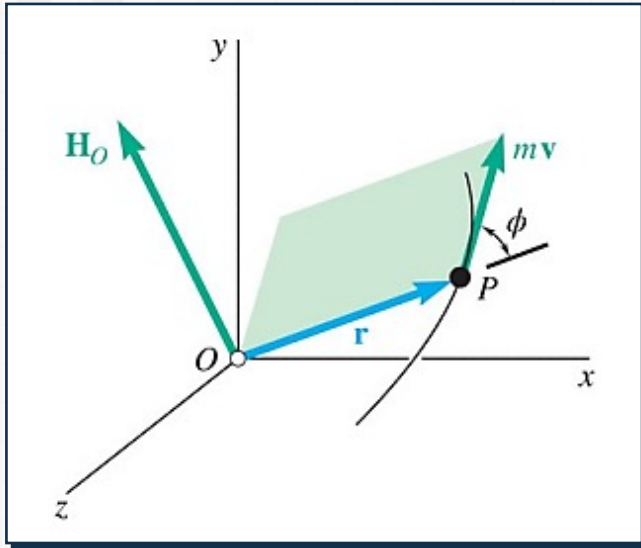
Angular Momentum of a Particle

$$\vec{H}_o = \vec{r} \times m\vec{v}$$

➡ moment of momentum or the angular momentum of the particle about O .

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Angular Momentum of a Particle



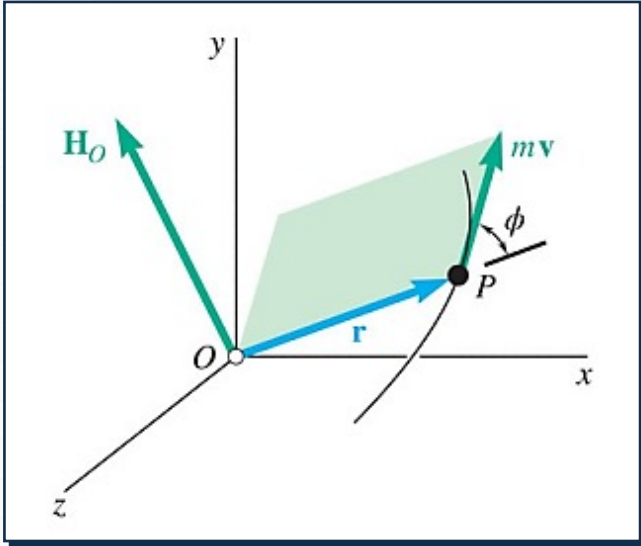
- $\vec{H}_O$ is perpendicular to plane containing $\vec{r}$ and $m\vec{v}$.

$$\begin{aligned} H_O &= r m v \sin \varphi \\ &= r m v_\theta \\ &= m r^2 \dot{\theta} \end{aligned}$$

$$\vec{H}_O = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ m v_x & m v_y & m v_z \end{vmatrix}$$

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Angular Momentum of a Particle

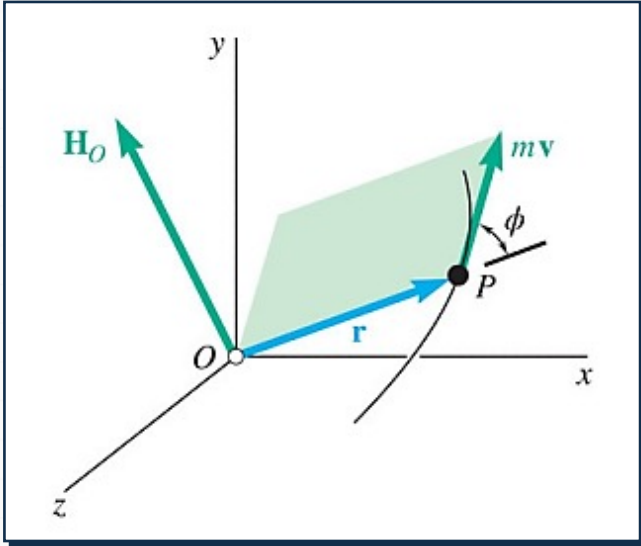


- Derivative of angular momentum with respect to time,

$$\begin{aligned}\dot{\vec{H}}_O &= \dot{\vec{r}} \times m\vec{v} + \vec{r} \times m\dot{\vec{v}} = \vec{v} \times m\vec{v} + \vec{r} \times m\vec{a} \\ &= \vec{r} \times \sum \vec{F} \\ &= \sum \vec{M}_O\end{aligned}$$

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Angular Momentum of a Particle

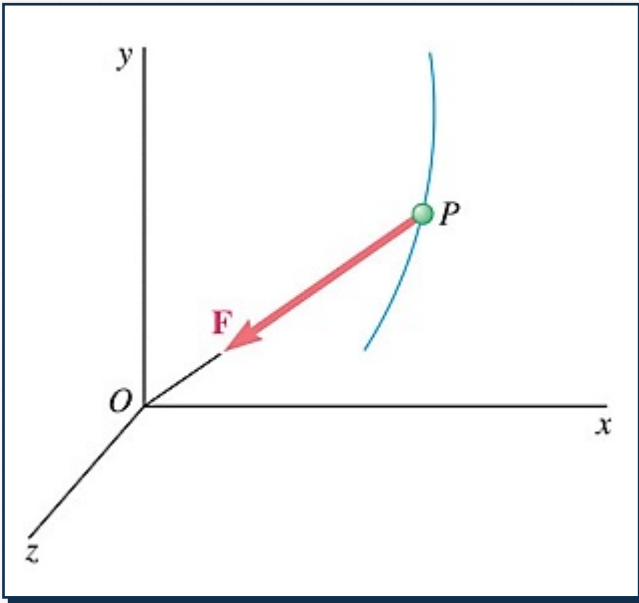


- the sum of the moments about O of the forces acting on the particle
 - It follows from Newton's second law that the sum of the moments about O of the forces acting on the particle is equal to the rate of change of the angular momentum of the particle about O .

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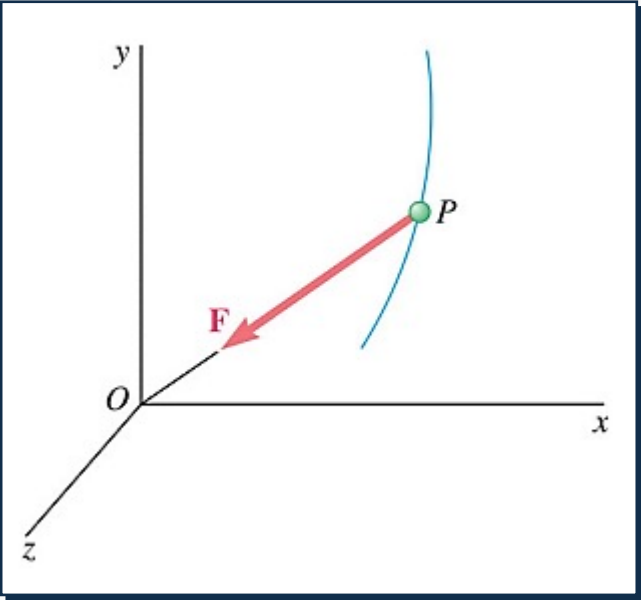
Conservation of Angular Momentum

When only force acting on particle is directed toward or away from a fixed point O , the particle is said to be moving under a central force.



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Conservation of Angular Momentum



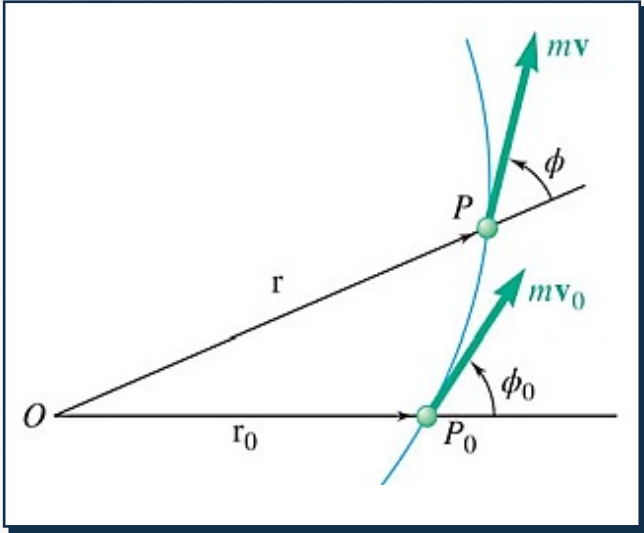
- Since the line of action of the central force passes through O ,

$$\sum \vec{M}_O = \dot{\vec{H}}_O = 0 \text{ and}$$

$$\vec{r} \times m\vec{v} = \vec{H}_O = \text{constant}$$

5 Motion under central force

Conservation of Angular Momentum



- Position vector and motion of particle are in a plane perpendicular to $\vec{H}_O$.
- Magnitude of angular momentum,

$$H_O = rmv \sin \phi = \text{constant} \\ = r_0 m v_0 \sin \phi_0$$

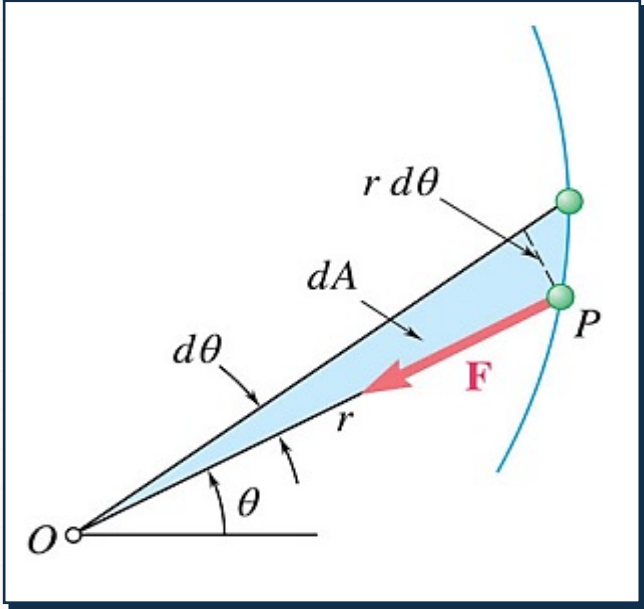
or

$$H_O = mr^2 \dot{\theta} = \text{constant}$$

$$\frac{H_O}{m} = r^2 \dot{\theta} = h = \frac{\text{angular momentum}}{\text{unit mass}}$$

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Conservation of Angular Momentum



- Radius vector OP sweeps infinitesimal area

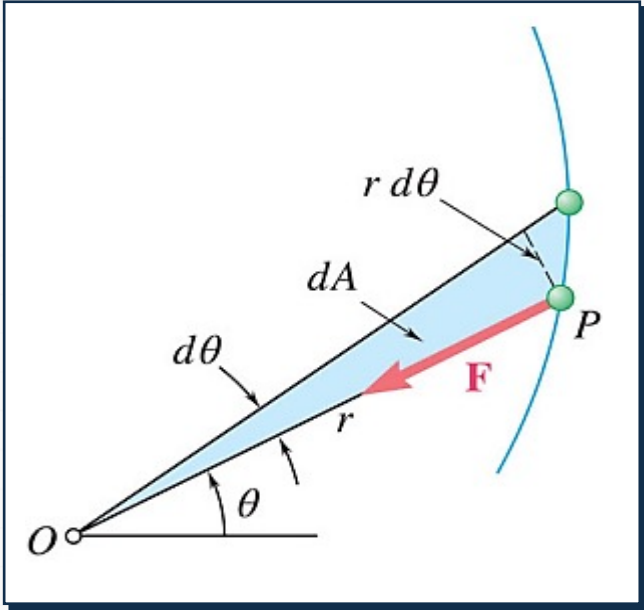
$$dA = \frac{1}{2}r^2 d\theta$$

- Define

$$\frac{dA}{dt} = \frac{1}{2}r^2 \frac{d\theta}{dt} = \frac{1}{2}r^2 \dot{\theta} = \text{areal velocity}$$

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Conservation of Angular Momentum



- Recall, for a body moving under a central force,

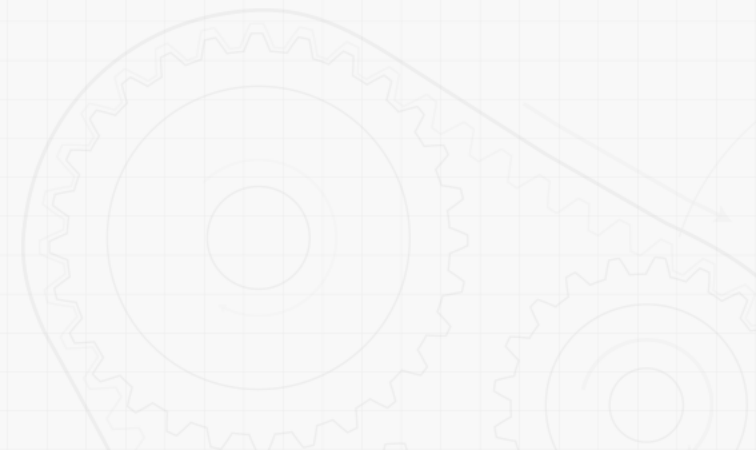
$$h = r^2 \dot{\theta} = \text{constant}$$

- When a particle moves under a central force, its areal velocity is constant.

5 Motion under central force

Newton's Law of Gravitation

Gravitational force exerted by the sun on a planet or by the earth on a satellite is an important example of gravitational force.



5 Motion under central force

Newton's Law of Gravitation

⚙ Newton's law of universal gravitation

- Two particles of mass M and m attract each other with equal and opposite force directed along the line connecting the particles,

$$F = G \frac{Mm}{r^2}$$

$G = \text{constant of gravitation}$

$$= 66.73 \times 10^{-12} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$$

5 Motion under central force

Newton's Law of Gravitation

⚙ Newton's law of universal gravitation

- For particle of mass m on the earth's surface,

$$W = m \frac{MG}{R^2} = mg \qquad g = 9.81 \frac{m}{s^2}$$

5 Motion under central force

Trajectory of a Particle Under a Central Force

⚙ For particle moving under central force directed towards force center,

$$m(\ddot{r} - r\dot{\theta}^2) = \sum F_r = -F \quad m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = \sum F_\theta = 0$$

- Second expression is equivalent to $r^2\dot{\theta} = h = \text{constant}$, from which,

$$\dot{\theta} = \frac{h}{r^2} \quad \text{and} \quad \ddot{r} = -\frac{h^2}{r^2} \frac{d^2}{d\theta^2} \left(\frac{1}{r} \right)$$

5 Motion under central force

Trajectory of a Particle Under a Central Force

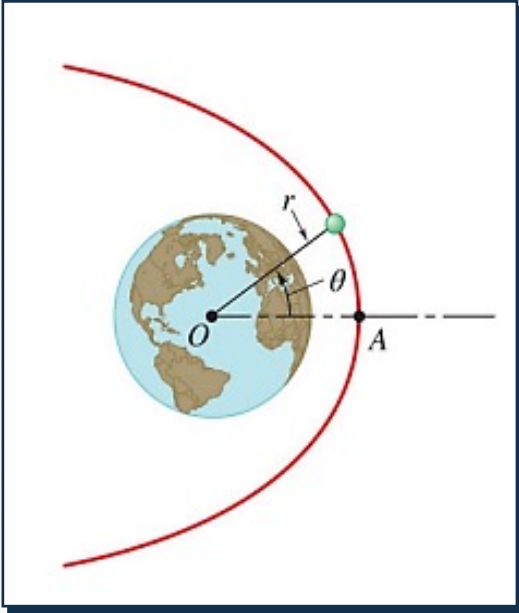
⚙ After substituting into the radial equation of motion and simplifying,

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{mh^2u^2} \quad \text{where } u = \frac{1}{r}$$

- If F is a known function of r or u , then particle trajectory may be found by integrating for $u=f(\theta)$, with constants of integration determined from initial conditions.

5 Motion under central force

Application to Space Mechanics



- Consider earth satellites subjected to only gravitational pull of the earth,

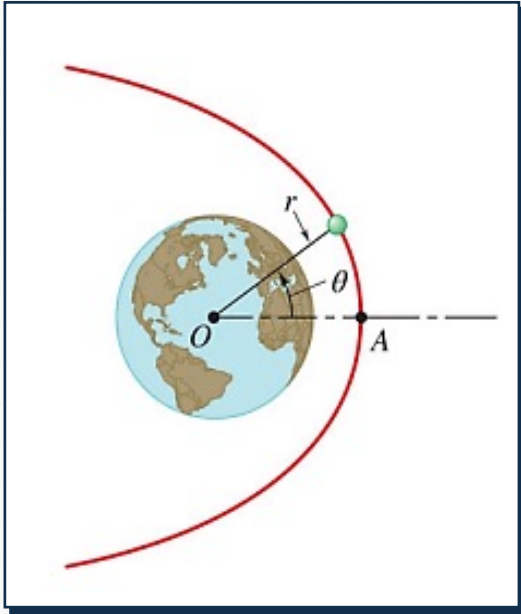
$$\frac{d^2u}{d\theta^2} + u = \frac{F}{mh^2u^2} \quad \text{where } u = \frac{1}{r}$$

$$F = \frac{GMm}{r^2} = GMmu^2$$

$$\frac{d^2u}{d\theta^2} + u = \frac{GM}{h^2} = \text{constant}$$

5 Motion under central force

Application to Space Mechanics



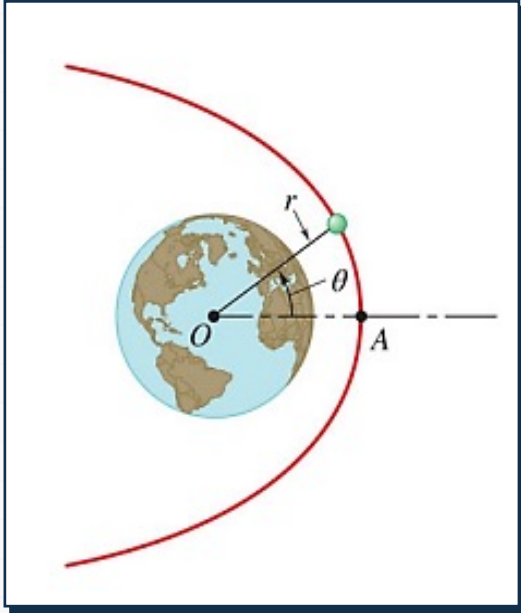
- Solution is equation of conic section,

$$u = \frac{1}{r} = \frac{GM}{h^2} (1 + \varepsilon \cos \theta) \quad \varepsilon = \frac{Ch^2}{GM} = \text{eccentricity}$$

- Origin, located at earth's center, is a focus of the conic section.

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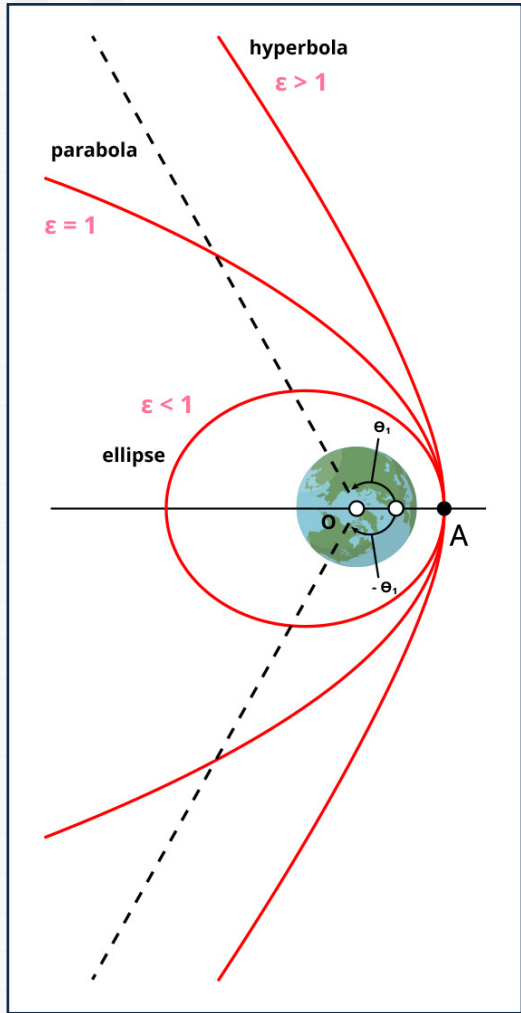
Application to Space Mechanics



- Trajectory may be ellipse, parabola, or hyperbola depending on value of eccentricity.

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Application to Space Mechanics

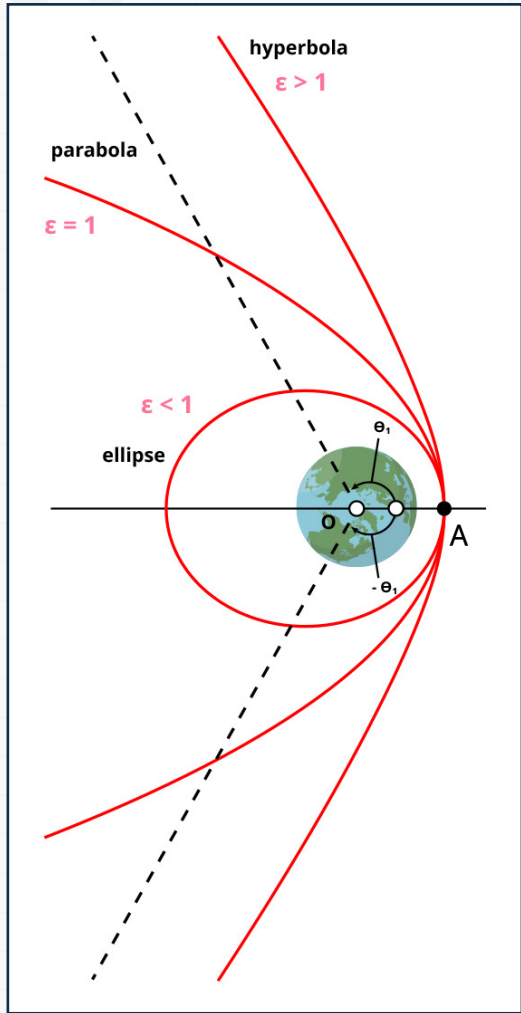


- Trajectory of earth satellite is defined by,

$$\frac{1}{r} = \frac{GM}{h^2} (1 + \epsilon \cos \theta) \quad \epsilon = \frac{Ch^2}{GM} = \text{eccentricity}$$

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Application to Space Mechanics



hyperbola

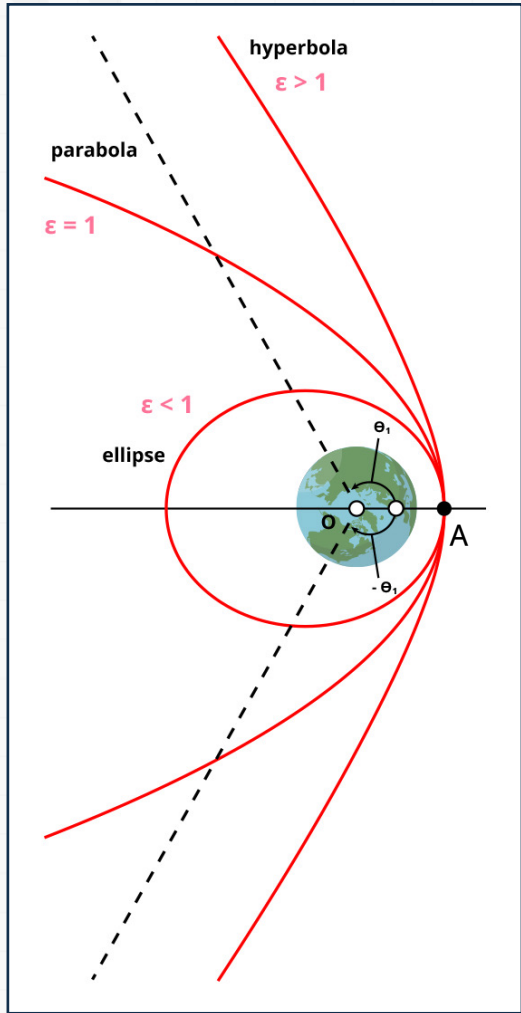
$$\epsilon > 1 \text{ or } C > GM/h^2$$

- The radius vector becomes infinite for,

$$1 + \epsilon \cos \theta_1 = 0 \quad \theta_1 = \pm \cos^{-1} \left(-\frac{1}{\epsilon} \right) = \pm \cos^{-1} \left(-\frac{GM}{Ch^2} \right)$$

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Application to Space Mechanics



parabola

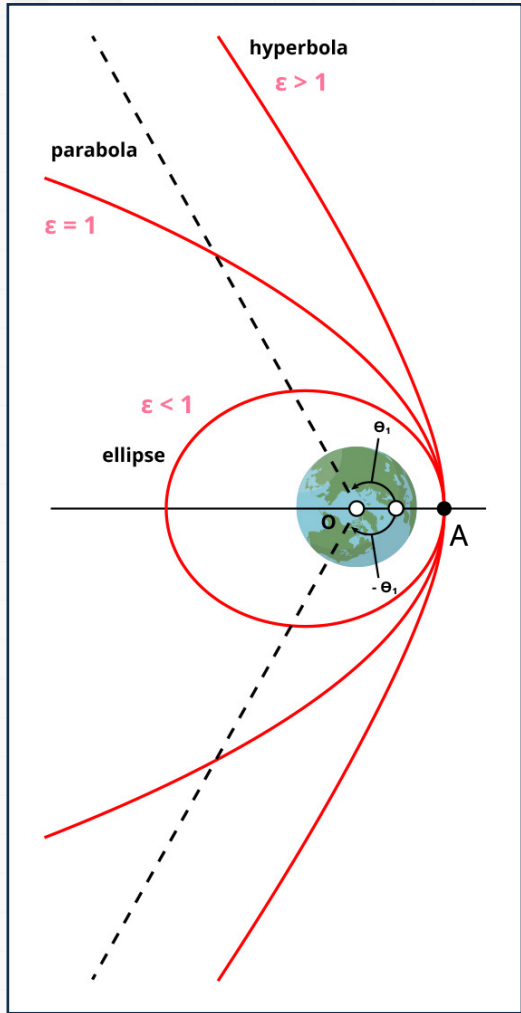
$$\epsilon = 1 \text{ or } C = GM/h^2$$

- The radius vector becomes infinite for,

$$1 + \cos \theta_2 = 0 \quad \theta_2 = 180^\circ$$

5 Motion under central force

Application to Space Mechanics



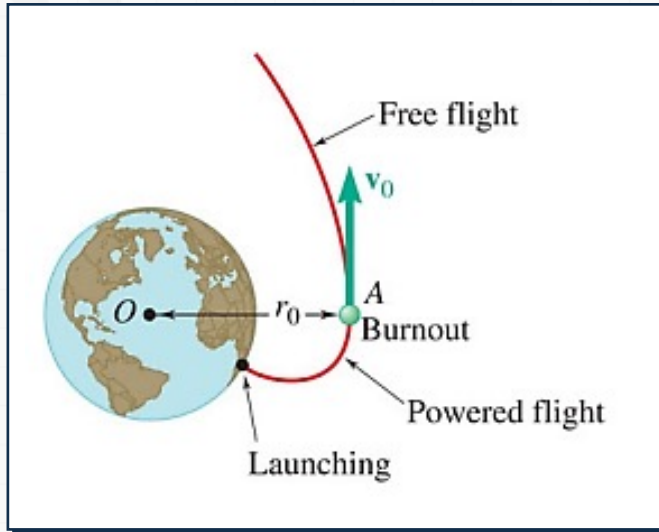
ellipse

$$\epsilon < 1 \text{ or } C < GM/h^2$$

- The radius vector is finite for q and is constant, that is a circle, for $\epsilon < 0$.

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Application to Space Mechanics



- Integration constant C

- Integration constant C is determined by conditions at beginning of free flight, $\theta = 0$,

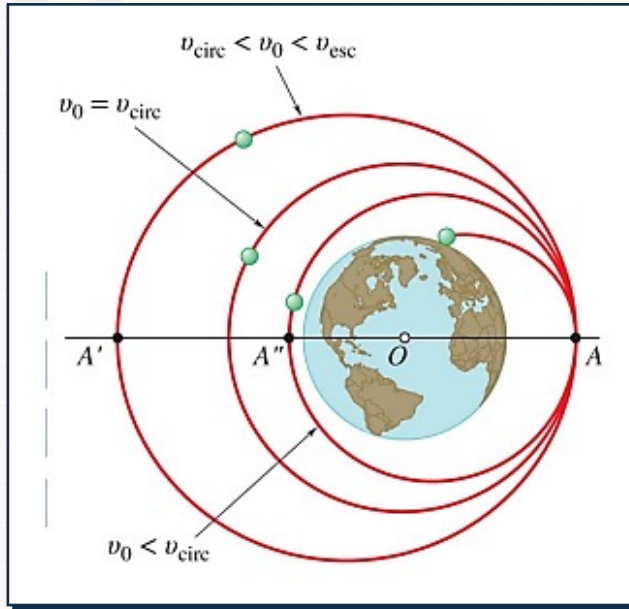
$$r = r_0,$$

$$\frac{1}{r_0} = \frac{GM}{h^2} \left(1 + \frac{Ch^2}{GM} \cos 0^\circ \right)$$

$$C = \frac{1}{r_0} - \frac{GM}{h^2} = \frac{1}{r_0} - \frac{GM}{(r_0 v_0)^2}$$

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Application to Space Mechanics



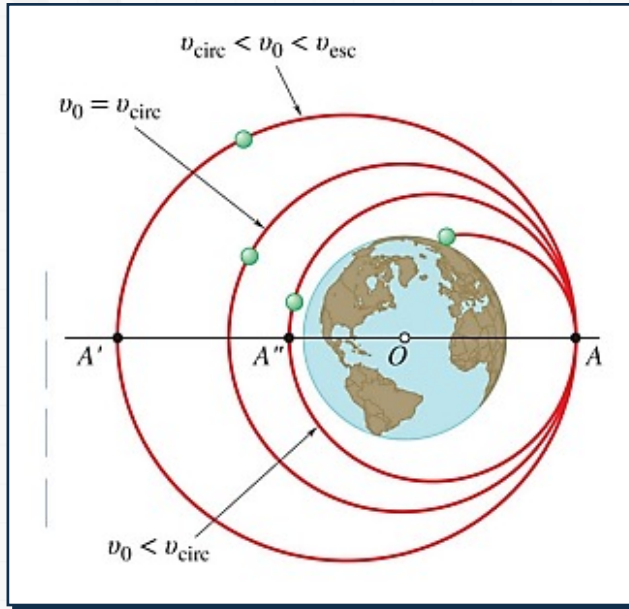
- Satellite escapes earth orbit for

$$\varepsilon \geq 1 \text{ or } C \geq GM/h^2 = GM/(r_0 v_0)^2$$

$$v_{\text{esc}} = v_0 = \sqrt{\frac{2GM}{r_0}}$$

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Application to Space Mechanics



- Trajectory is elliptic for $v_0 < v_{\text{esc}}$ and becomes circular for $\varepsilon = 0$ or $C = 0$,

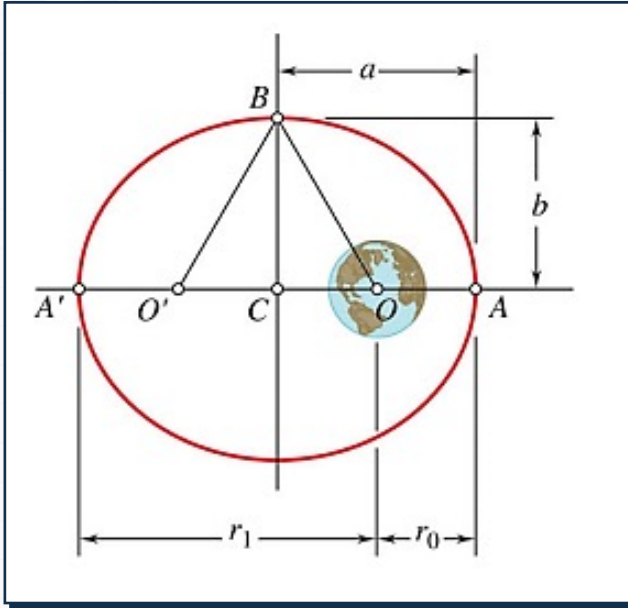
$$v_{\text{circ}} = \sqrt{\frac{GM}{r_0}}$$

Application to Space Mechanics

- $$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \frac{1}{2}h = \text{constant}$$

5 Motion under central force

Application to Space Mechanics



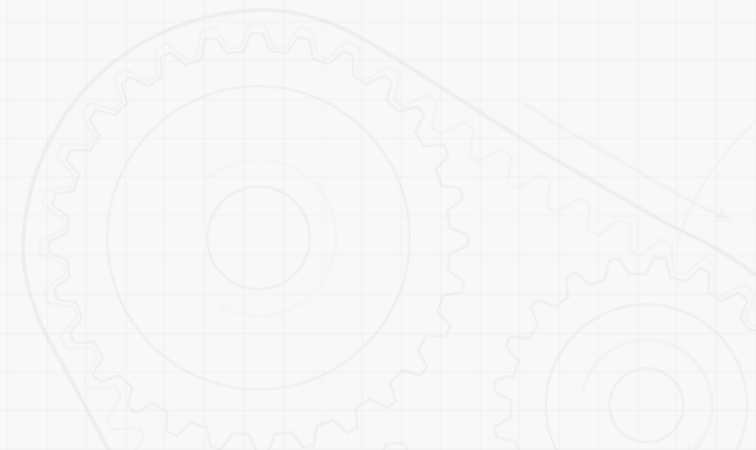
- Periodic time or time required for a satellite to complete an orbit is equal to area within the orbit divided by *areal velocity*,

$$\tau = \frac{\pi ab}{h/2} = \frac{2\pi ab}{h} \quad \text{where} \quad a = \frac{1}{2}(r_0 + r_1) \\ b = \sqrt{r_0 r_1}$$

5 Motion under central force

Kepler's Laws of Planetary Motion

Results obtained for trajectories of satellites around earth may also be applied to trajectories of planets around the sun.



5 Motion under central force

Kepler's Laws of Planetary Motion

Properties of planetary orbits around the sun were determined astronomical observations by Johann Kepler (1571 to 1630) before Newton had developed his fundamental theory.

Each planet describes an ellipse, with the sun located at one of its foci.

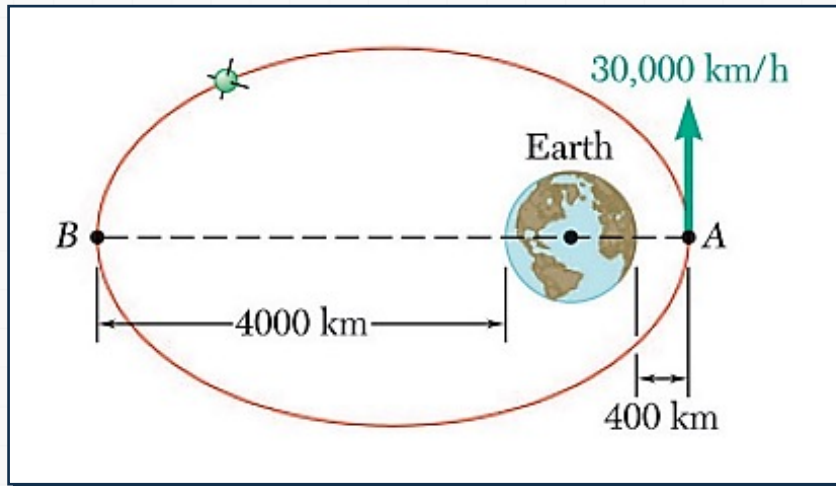
The radius vector drawn from the sun to a planet sweeps equal areas in equal times.

The squares of the periodic times of the planets are proportional to the cubes of the semimajor axes of their orbits.

5 Motion under central force

Sample Problem 1

⚙️ A satellite is launched in a direction parallel to the surface of the earth with a velocity of $30,000 \text{ km/h}$ from an altitude of 400 km .



• Determine

- The velocity of the satellite as it reaches its maximum altitude of 4000 km .
- The radius of the earth is 6370 km .

5 Motion under central force

Sample Problem 1

⚙️ A satellite is launched in a direction parallel to the surface of the earth with a velocity of 30,000 km/h from an altitude of 400 km.

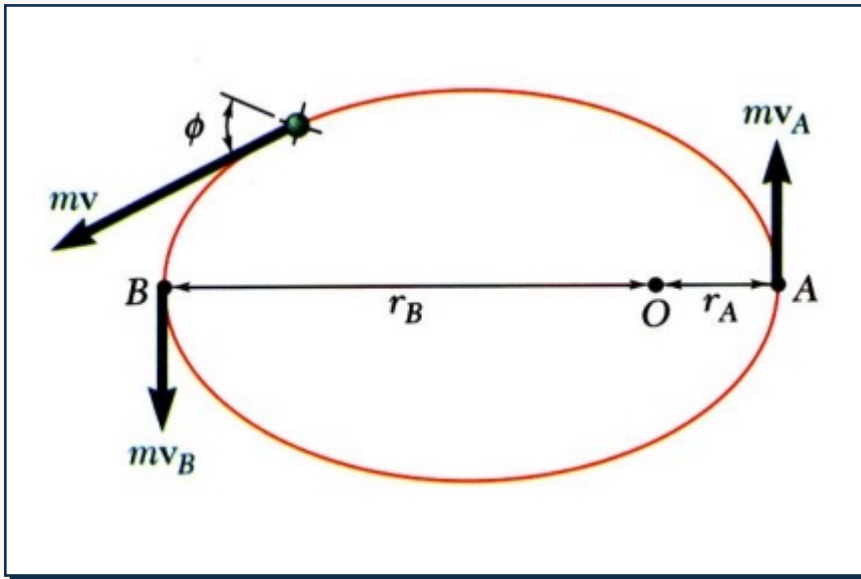
• Strategy

- Since the satellite is moving under a central force, its angular momentum is constant.
- Equate the angular momentum at A and B and solve for the velocity at B .

5 Motion under central force

Sample Problem 1

⚙ Modeling and Analysis



$$rmv \sin \phi = H_o = \text{constant}$$

$$r_A m v_A = r_B m v_B$$

$$v_B = v_A \frac{r_A}{r_B}$$

$$= (30,000 \text{ km/h}) \frac{(6370 + 400) \text{ km}}{(6370 + 4000) \text{ km}}$$

$$v_B = 19,590 \text{ km/h}$$

5 Motion under central force

Sample Problem 1

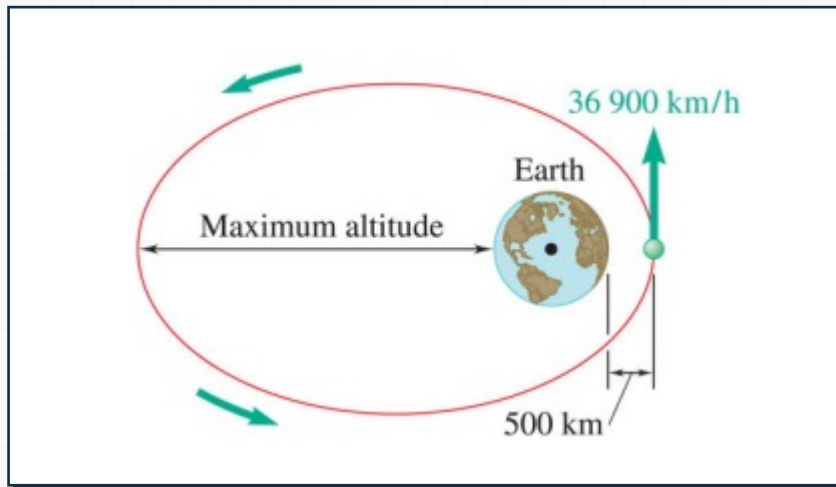
⚙️ Reflect and Think

Note that in order to increase velocity, a spacecraft often applies thrusters to push it closer to the earth. This central force means the spacecraft's angular momentum remains constant, its radial distance r decreases, and its velocity v increases.

5 Motion under central force

Sample Problem 2

⚙️ A satellite is launched in a direction parallel to the surface of the earth with a velocity of $36,900 \text{ km/h}$ at an altitude of 500 km .



• Determine

- a) the maximum altitude reached by the satellite
- b) the periodic time of the satellite

5 Motion under central force

Sample Problem 2

⚙️ A satellite is launched in a direction parallel to the surface of the earth with a velocity of 36,900 km/h at an altitude of 500 km.

• Strategy

- Trajectory of the satellite is described by,

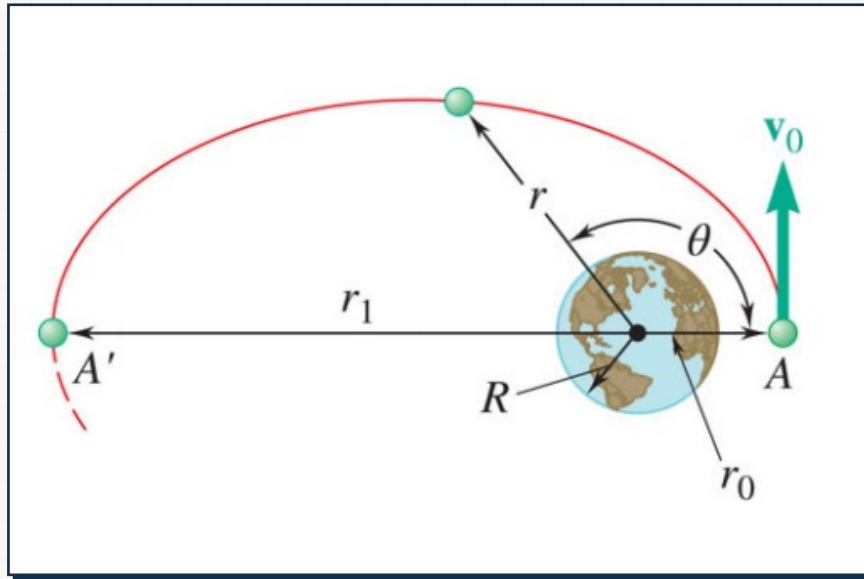
$$\frac{1}{r} = \frac{GM}{h^2} + C \cos \theta$$

- Evaluate C using the initial conditions at $\theta = 0$.
- Determine the maximum altitude by finding r at $\theta = 180^\circ$.
- With the altitudes at the perigee and apogee known, the periodic time can be evaluated.

5 Motion under central force

Sample Problem 2

⚙ Modeling and Analysis



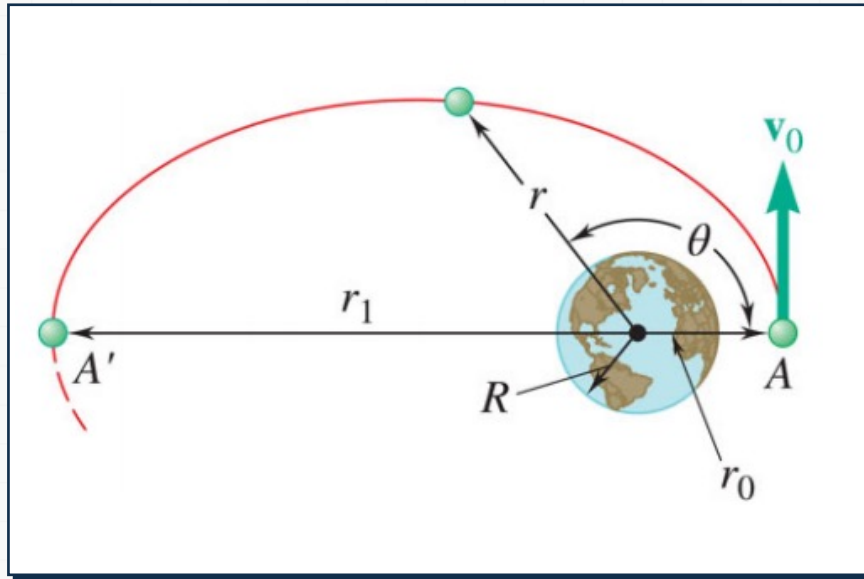
- Trajectory of the satellite is described by,

$$\frac{1}{r} = \frac{GM}{h^2} + C \cos \theta$$

5 Motion under central force

Sample Problem 2

⚙ Modeling and Analysis



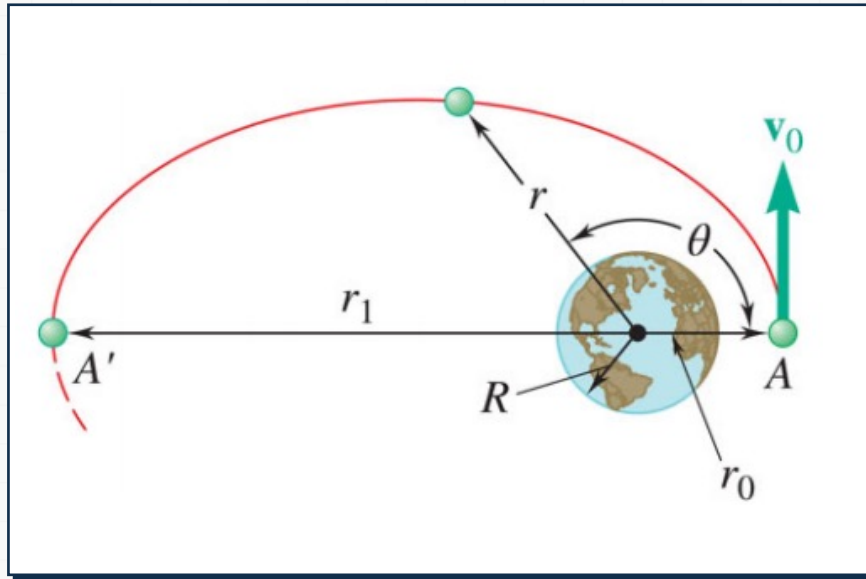
- Evaluate C using the initial conditions at $\theta = 0$.

$$\begin{aligned} C &= \frac{1}{r_0} - \frac{GM}{h^2} \\ &= \frac{1}{6.87 \times 10^6 \text{ m}} - \frac{398 \times 10^{12} \text{ m}^3/\text{s}^2}{(70.4 \text{ m}^2/\text{s})^2} \\ &= 65.3 \times 10^{-9} \text{ m}^{-1} \end{aligned}$$

5 Motion under central force

Sample Problem 2

⚙ Modeling and Analysis



$$r_0 = (6370 + 500)km$$

$$= 6.87 \times 10^6 m$$

$$v_0 = 36900 \frac{km}{h} \times \frac{1000m/km}{3600s/h}$$

$$= 10.25 \times 10^3 m/s$$

$$h = v_0 r_0 = (6.87 \times 10^6 m)(10.25 \times 10^3 m/s)$$

$$= 70.4 \times 10^9 m^2/s$$

$$GM = gR^2 = (9.81 m/s^2)(6.37 \times 10^6 m)^2$$

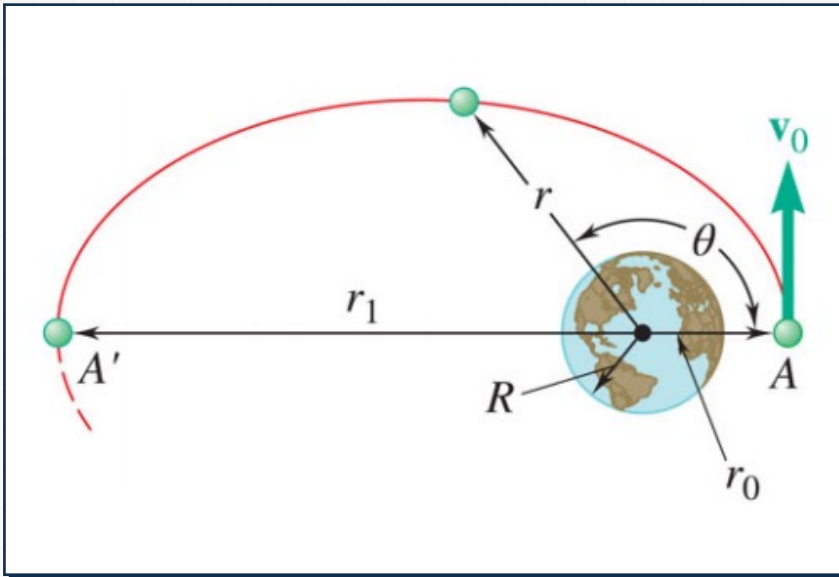
$$= 398 \times 10^{12} m^3/s^2$$

5 Motion under central force

Sample Problem 2

⚙ Modeling and Analysis

- Determine the maximum altitude by finding r_1 at $\theta = 180^\circ$



$$\frac{1}{r_1} = \frac{GM}{h^2} - C = \frac{398 \times 10^{12} \text{ m}^3/\text{s}^2}{(70.4 \text{ m}^2/\text{s})^2} - 65.3 \times 10^{-9} \frac{1}{\text{m}}$$

$$r_1 = 66.7 \times 10^6 \text{ m} = 66700 \text{ km}$$

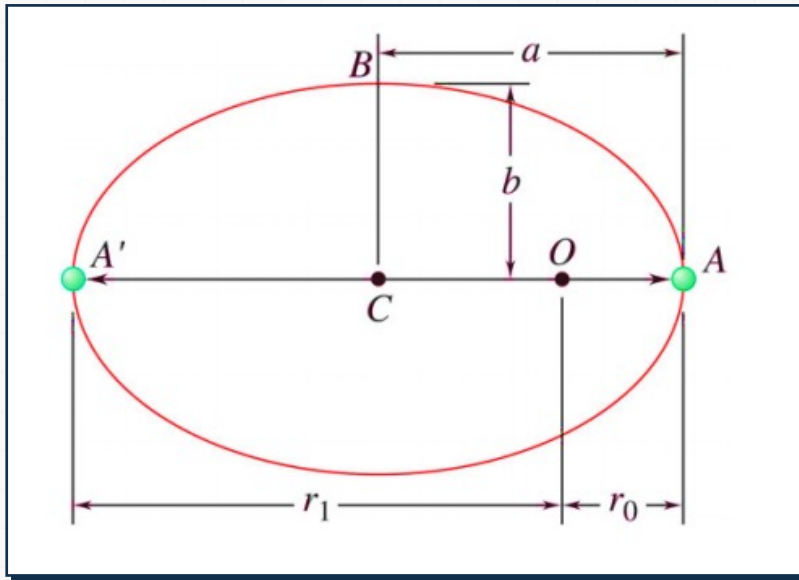
$$\text{max altitude} = (66700 - 6370) \text{ km} = 60300 \text{ km}$$

5 Motion under central force

Sample Problem 2

⚙ Modeling and Analysis

- With the altitudes at the perigee and apogee known, the periodic time can be evaluated.



$$a = \frac{1}{2}(r_0 + r_1) = \frac{1}{2}(6.87 + 66.7) \times 10^6 \text{ m} = 36.8 \times 10^6 \text{ m}$$

$$b = \sqrt{r_0 r_1} = \sqrt{6.87 \times 66.7} \times 10^6 \text{ m} = 21.4 \times 10^6 \text{ m}$$

$$\tau = \frac{2\pi ab}{h} = \frac{2\pi(36.8 \times 10^6 \text{ m})(21.4 \times 10^6 \text{ m})}{70.4 \times 10^9 \text{ m}^2/\text{s}}$$

$$\tau = 70.3 \times 10^3 \text{ s} = 19 \text{ h } 31 \text{ min}$$

5 Motion under central force

Sample Problem 2

⚙️ Reflect and Think

The satellite takes less than one day to travel over 60,000 km from the earth and back. In this problem, you started with Equation 12.37, but it is important to remember that this formula was the solution to a differential equation that was derived using Newton's second law.

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, Chapter 12